SINGAPORE SCHOOL

CONTINUAL ASSESSMENT 1

Secondary 2

Computer Science (Part 1)

2026-27

Date: ___ / ___ / ___ **Name: _____**

Instructions for students:

- **Start answering the questions on time.**
- **Follow all the instructions carefully.**
- **Answer all questions.**
- **Revise**

CA 1 Marks (Part – 1)	**25**
CW	**05**
Total Marks Obtained	**30**

1. Multiple Choice Questions: Tick (✓) the correct option. **(1 x 5 = 5)**

(a) Why do computers process all data in binary (0s and 1s)?
[ ] i) Circuits use transistors with two states: ON (1) and OFF (0) [ ] ii) Binary has 16 digits [ ] iii) It uses no electricity

(b) How many bits are used in the Standard ASCII character set?
[ ] i) 7 bits [ ] ii) 16 bits [ ] iii) 32 bits

(c) What is the unit of measurement for sound Sample Rate?
[ ] i) Pixels [ ] ii) Hertz (Hz) [ ] iii) Bytes

(d) An image with an 8-bit colour depth can display how many unique colours?
[ ] i) 16 colours [ ] ii) 256 colours [ ] iii) 65,536 colours

(e) In HTML hexadecimal colour codes, which code represents pure Black?
[ ] i) #000000 [ ] ii) #FFFFFF [ ] iii) #FF0000

2. Fill in the blanks using the correct words from the Word Bank below. **(1 x 5 = 5)**

Word Bank: [Pixel | ADC | Unicode | MAC Address | Metadata]

(a) The _____ character set uses 16 or 32 bits and supports all world languages and emojis.

(b) An _____ (Analogue-to-Digital Converter) converts analogue sound waves into digital binary data.

(c) The smallest single dot or element of a bitmap image is called a _____.

(d) Extra information stored inside an image file (such as dimensions and colour depth) is called _____.

(e) A unique 48-bit (12-hex-digit) hardware address hardcoded on a Network Interface Card (NIC) is a _____.

3. State whether each of the following statements is True or False. **(1 x 5 = 5)**

(a) One hexadecimal digit replaces exactly 4 binary bits (1 nibble). [ ]

(b) In a MAC Address, the first 6 hexadecimal digits represent the Manufacturer Code (OUI). [ ]

(c) IPv6 addresses are 128-bit network addresses written as 8 groups of 4 hexadecimal digits. [ ]

(d) Lossy compression should be used to compress computer program executable files (program.exe). [ ]

(e) A 4-bit colour depth image can display a maximum of 16 unique colours. [ ]

4. Number Systems & Hexadecimal Applications (From CT 1 Revision Guide PDF) **(3 + 2 = 5)**

(a) Convert the following numbers accurately: **[3 marks]**

i) Binary **1010 1110** into Denary (Decimal): _____

ii) Binary **0111 1101 0100** into Hexadecimal: _____

iii) Hexadecimal **5B8** into 12-bit Binary: _____

(b) State **two** reasons why computer scientists prefer using Hexadecimal instead of pure Binary. **[2 marks]**

1.
_____

2.
_____

5. Data Compression, Run-Length Encoding (RLE) & Text Size (From Topic 1.2 PDF) (3 + 1 + 1 = 5)

(a) Tick (✓) one box in each row to show whether **Lossy** or **Lossless** compression is most suitable: **[3 marks]**

File Description	Lossy Compression	Lossless Compression
A high-resolution photograph to upload on social media (JPEG)	[ ]	[ ]
A computer program executable file (program.exe)	[ ]	[ ]
A music track for an online music streaming app (MP3)	[ ]	[ ]

(b) Compress the following row of 16 pixels using **Run-Length Encoding (RLE)**: **[1 mark]**
`W W W W W W  B B B  W W W W  B B B`

RLE Representation:
_____

(c) A short story document contains **4,096 characters** encoded in 8-bit Extended ASCII (1 Byte per character). State its file size in **Bytes** and in **Kibibytes (KiB)**: **[1 mark]**

Size in Bytes: _____ | Size in KiB: _____